

MySQL Assignment 2 – Querying Data (Documentation)

★ From the previous assignment all the database has been fetch.

★ By Using the provided **DML insert codes** to populate the tables from here: Employee Data

```
INSERT INTO departments (department_id, department_name) VALUES
(1, 'Software Development'),
(2, 'Marketing'),
(3, 'Data Science'),
(4, 'Human Resources'),
(5, 'Product Management'),
(6, 'Content Creation'),
(7, 'Finance'),
(8, 'Design'),
(9, 'Research and Development'),
(10, 'Customer Support'),
(11, 'Business Development'),
(12, 'IT'),
(13, 'Operations');
```

- **INSERT INTO location (location) VALUES**
('Chennai'),
('Bangalore'),
('Hyderabad'),
('Pune');

- **INSERT INTO employees (employee_id, employee_name, gender, age, hire_date, designation, department_id, location_id, salary) VALUES**
(5001, 'Vihaan Singh', 'M', 27, '2015-01-20', 'Data Analyst', 3, 4, 60000),
(5002, 'Reyansh Singh', 'M', 31, '2015-03-10', 'Network Engineer', 12, 1, 80000),
(5003, 'Aaradhya Iyer', 'F', 26, '2015-05-20', 'Customer Support Executive', 10, 2, 45000),
(5004, 'Kiara Malhotra', 'F', 29, '2015-07-05', NULL, 8, 3, 70000),
(5005, 'Anvi Chaudhary', 'F', 25, '2015-09-11', 'Business Development Executive', 11, 1, 55000),
(5006, 'Dhruv Shetty', 'M', 28, '2015-11-20', 'UI Developer', 8, 2, 65000),
(5007, 'Anushka Singh', 'F', 32, '2016-01-15', 'Marketing Manager', 2, 3, 90000),
(5008, 'Diya Jha', 'F', 27, '2016-03-05', 'Graphic Designer', 8, 4, 70000),
(5009, 'Kiaan Desai', 'M', 30, '2016-05-20', 'Sales Executive', 11, 3, 55000),
(5010, 'Atharv Yadav', 'M', 29, '2016-07-10', 'Systems Administrator', 12, 4, 80000),
(5011, 'Saanvi Patel', 'F', 28, '2016-09-20', 'Marketing Analyst', 2, 1, 60000),
(5012, 'Myra Verma', 'F', 26, '2016-11-05', 'Operations Manager', 13, 2, 95000),
(5013, 'Arnav Rao', 'M', 33, '2017-01-20', 'Customer Success Manager', 10, 3, 75000),
(5014, 'Vihaan Mohan', 'M', 30, '2017-03-10', 'Supply Chain Analyst', 10, 2, 60000),
(5015, 'Ishaan Kumar', 'M', 27, '2017-05-20', 'Financial Analyst', 7, 1, 85000),
(5016, 'Zoya Khan', 'F', 31, '2017-07-05', 'Legal Counsel', 4, 4, 100000),
(5017, 'Kabir Nair', 'M', 28, '2017-09-11', 'IT Support Specialist', 12, 2, 80000),
(5018, 'Ishan Mishra', 'M', 25, '2017-11-20', 'Research Scientist', 9, 3, 75000),

```
(5019, 'Ishika Patel', 'F', 29, '2018-01-15', 'Talent Acquisition Specialist', 4, 4, 55000),
(5020, 'Aarav Nair', 'M', 32, '2018-03-05', 'Software Engineer', 1, 1, 90000),
(5021, 'Advik Kapoor', 'M', 26, '2018-05-20', 'Finance Analyst', 7, 3, 85000),
(5022, 'Aadhya Iyengar', 'F', 28, '2018-07-10', 'HR Specialist', 4, 4, 60000),
(5023, 'Anika Paul', 'F', 30, '2018-09-20', 'Public Relations Specialist', 2, 2, 70000),
(5024, 'Aryan Shetty', 'M', 27, '2018-11-05', 'Product Manager', 5, 1, 95000),
(5025, 'Avni Iyengar', 'F', 31, '2019-01-20', 'Data Scientist', 3, 4, 100000),
(5026, 'Vivaan Singh', 'M', 29, '2019-03-10', 'Business Analyst', 3, 2, 75000),
(5027, 'Ananya Paul', 'F', 32, '2019-05-20', 'Content Writer', 6, 3, 60000),
(5028, 'Anaya Kapoor', 'F', 26, '2019-07-05', 'Event Coordinator', 6, 1, 60000),
(5029, 'Arjun Kumar', 'M', 33, '2019-09-11', 'Quality Assurance Analyst', 12, 2, 80000),
(5030, 'Sara Iyer', 'F', 28, '2019-11-20', 'Project Manager', 5, 1, 90000);
```

All the data has been inserted into the database.

1. Distinct Values:

- a query to retrieve distinct salaries from the Employees table.

```
87 • SELECT DISTINCT salary
88     FROM Employees;
89
```

salary
60000.00
80000.00
45000.00
70000.00
55000.00
65000.00
90000.00
95000.00
75000.00
85000.00
100000.00

2. Alias (AS):

- Provide aliases for the "age" and "salary" columns as "Employee_Age" and "Employee_Salary", respectively.

```
92 • SELECT age AS Employee_Age,
93          salary AS Employee_Salary
94     FROM Employees;
95
```

Employee_Age	Employee_Salary
29	70000.00
25	55000.00
28	65000.00
32	90000.00
27	70000.00
30	55000.00
29	80000.00
28	60000.00
26	95000.00
33	75000.00
30	60000.00
27	85000.00
31	100000.00
28	80000.00
25	75000.00
29	55000.00
32	90000.00
26	85000.00
28	60000.00
30	70000.00
27	95000.00
31	100000.00
29	75000.00
32	60000.00
26	60000.00

3. Where Clause & Operators:

- Retrieve employees with a salary greater than ₹50000 and hired before 2016-01-01.

```
96 • SELECT *
97 FROM Employees
98 WHERE salary > 50000
99 AND hire_date < '2016-01-01';
100
```

employee_id	employee_name	gender	age	hire_date	designation	department_id	location_id	salary
5001	Vihaan Singh	M	27	2015-01-20	Data Analyst	3	4	60000.00
5002	Reyansh Singh	M	31	2015-03-10	Network Engineer	12	1	80000.00
5004	Kiara Malhotra	F	29	2015-07-05	NULL	8	3	70000.00
5005	Anvi Chaudhary	F	25	2015-09-11	Business Development Executive	11	1	55000.00
5006	Dhruv Shetty	M	28	2015-11-20	UI Developer	8	2	65000.00
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

- Find the employee whose designation is missing and fill it with "Data Scientist".

```
110 • UPDATE Employees
111 SET designation = 'Data Scientist'
112 WHERE designation IS NULL and employee_id=5004;
113
114 • select * from employees;
115 • SELECT *
```

employee_id	employee_name	gender	age	hire_date	designation	department_id	location_id	salary
5001	Vihaan Singh	M	27	2015-01-20	Data Analyst	3	4	60000.00
5002	Reyansh Singh	M	31	2015-03-10	Network Engineer	12	1	80000.00
5003	Aaradhya Iyer	F	26	2015-05-20	Customer Support Executive	10	2	45000.00
5004	Kiara Malhotra	F	29	2015-07-05	Data Scientist	8	3	70000.00
5005	Anvi Chaudhary	F	25	2015-09-11	Business Development Executive	11	1	55000.00
5006	Dhruv Shetty	M	28	2015-11-20	UI Developer	8	2	65000.00
5007	Anushka Singh	F	32	2016-01-15	Marketing Manager	2	3	90000.00
5008	Diya Jha	F	27	2016-03-05	Graphic Designer	8	4	70000.00
5009	Kiran Patel	M	30	2016-05-20	Sales Executive	11	3	55000.00

Sorting and Grouping Data:

1. ORDER BY:

```
107 • SELECT *
108 FROM Employees
109 ORDER BY department_id ASC, salary DESC;
```

employee_id	employee_name	gender	age	hire_date	designation	department_id	location_id	salary
5020	Aarav Nair	M	32	2018-03-05	Software Engineer	1	1	90000.00
5007	Anushka Singh	F	32	2016-01-15	Marketing Manager	2	3	90000.00
5023	Anika Paul	F	30	2018-09-20	Public Relations Specialist	2	2	70000.00
5011	Saanvi Patel	F	28	2016-09-20	Marketing Analyst	2	1	60000.00
5025	Avni Iyengar	F	31	2019-01-20	Data Scientist	3	4	100000.00
5026	Vivaan Singh	M	29	2019-03-10	Business Analyst	3	2	75000.00
5001	Vihaan Singh	M	27	2015-01-20	Data Analyst	3	4	60000.00
5016	Zoya Khan	F	31	2017-07-05	Legal Counsel	4	4	100000.00
5022	Aadhya Iyengar	F	28	2018-07-10	HR Specialist	4	4	60000.00
5019	Ishika Patel	F	29	2018-01-15	Talent Acquisition Specialist	4	4	55000.00
5024	Aryan Shetty	M	27	2018-11-05	Product Manager	5	1	95000.00
5030	Sara Iyer	F	28	2019-11-20	Project Manager	5	1	90000.00
5027	Ananya Paul	F	32	2019-05-20	Content Writer	6	3	60000.00
5028	Anaya Kapoor	F	26	2019-07-05	Event Coordinator	6	1	60000.00
5015	Ishaan Kumar	M	27	2017-05-20	Financial Analyst	7	1	85000.00
5021	Advik Kapoor	M	26	2018-05-20	Finance Analyst	7	3	85000.00
5004	Kiara Malhotra	F	29	2015-07-05	NULL	8	3	70000.00
5008	Diya Jha	F	27	2016-03-05	Graphic Designer	8	4	70000.00
5006	Dhruv Shetty	M	28	2015-11-20	UI Developer	8	2	65000.00
5018	Ishan Mishra	M	25	2017-11-20	Research Scientist	9	3	75000.00
5013	Arnav Rao	M	33	2017-01-20	Customer Success Manager	10	3	75000.00

2. LIMIT:

```
111 • SELECT *
112 FROM Employees
113 WHERE YEAR(hire_date) = 2018
114 LIMIT 5;
```

	employee_id	employee_name	gender	age	hire_date	designation	department_id	location_id	salary
▶	5019	Ishika Patel	F	29	2018-01-15	Talent Acquisition Specialist	4	4	55000.00
	5020	Aarav Nair	M	32	2018-03-05	Software Engineer	1	1	90000.00
	5021	Advik Kapoor	M	26	2018-05-20	Finance Analyst	7	3	85000.00
	5022	Aadhya Iyengar	F	28	2018-07-10	HR Specialist	4	4	60000.00
	5023	Anika Paul	F	30	2018-09-20	Public Relations Specialist	2	2	70000.00
•	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

3. Aggregate Functions:

- Sum of salaries in Finance:

```
132 • SELECT SUM(salary) AS Total_Salary
133 FROM Employees
134 WHERE department_id= 'Finance';
135
```

	Total_Salary
▶	NULL

- Find the minimum age among all employees

```
133 • SELECT MIN(age)
134 FROM Employees;
135
```

	MIN(age)
▶	25

4. GROUP BY:

- List the maximum salary for each location.

```
138 • SELECT location_id, MAX(salary)
139 FROM Employees
140 GROUP BY location_id;
141
```

	location_id	MAX(salary)
▶	1	95000.00
	2	95000.00
	3	90000.00
	4	100000.00

- Calculate the average salary for each designation containing the word 'Analyst'.

```

144 • SELECT designation, AVG(salary)
145 FROM Employees
146 WHERE designation LIKE '%Analyst%'
147 GROUP BY designation;
148

```

designation	AVG(salary)
Data Analyst	60000.000000
Marketing Analyst	60000.000000
Supply Chain Analyst	60000.000000
Financial Analyst	85000.000000
Finance Analyst	85000.000000
Business Analyst	75000.000000
Quality Assurance Analyst	80000.000000

5. HAVING:

- Find departments with less than 3 employees.

```

149 • SELECT department_id, COUNT(*) AS emp_count
150 FROM Employees
151 GROUP BY department_id
152 HAVING COUNT(*) < 3;
153

```

department_id	emp_count
1	1
5	2
6	2
7	2
9	1
11	2
13	1

- Find locations with female employees whose average age is below 30.

```

155 • SELECT location_id, AVG(age) AS Avg_Age
156 FROM Employees
157 WHERE gender = 'F'
158 GROUP BY location_id
159 HAVING AVG(age) < 30;
160

```

location_id	Avg_Age
1	26.7500
2	27.3333
4	29.2000

Joins:

1. Inner Join:

- List employee names, their designations, and department names where employees are assigned to a department.

```

162 • SELECT e.employee_name, e.designation, d.department_name
163 FROM Employees e
164 INNER JOIN Departments d
165 ON e.department_id = d.department_id;
166

```

employee_name	designation	department_name
Anvi Chaudhary	Business Development Executive	Business Development
Klaan Desai	Sales Executive	Business Development
Ananya Paul	Content Writer	Content Creation
Anaya Kapoor	Event Coordinator	Content Creation
Aaradhya Iyer	Customer Support Executive	Customer Support
Amav Rao	Customer Success Manager	Customer Support
Vhaan Mohan	Supply Chain Analyst	Customer Support
Vhaan Singh	Data Analyst	Data Science

2. Left Join:

☉ List all departments along with the total number of employees in each department, including departments with no employees.

```
167 • SELECT d.department_name, COUNT(e.employee_id) AS Employee_Count
168 FROM Departments d
169 LEFT JOIN Employees e
170 ON d.department_id = e.department_id
171 GROUP BY d.department_name;
172
```

department_name	Employee_Count
Business Development	2
Content Creation	2
Customer Support	3
Data Science	3
Design	3
Finance	2
Human Resources	3
IT	4
Marketing	2

3. Right Join:

☉ Display all locations along with the names of employees assigned to each location. If no employees are assigned to a location, display NULL for employee name.

```
173 • SELECT l.location_id, e.employee_name AS Employee_Name
174 FROM Employees e
175 RIGHT JOIN Location l
176 ON e.location_id = l.location_id;
177
```

location_id	Employee_Name
2	Aaradhya Iyer
2	Dhruv Shetty
2	Myra Verma
2	Vihaan Mohan
2	Kabir Nair
2	Anika Paul
2	Vivaan Singh
2	Arjun Kumar
1	Davaneh Singh

Conclusion

This assignment strengthened the understanding of SQL querying by applying concepts such as filtering, sorting, aggregation, grouping, and joins. It demonstrated how raw data can be refined, summarized, and connected across tables to generate meaningful insights. Overall, the tasks highlighted SQL's role as a powerful tool for data analysis and decision-making.